

The Supply Edge: A Mass-Conserving Rule Without Tuned Radii Under a Vacuum-Tied MOND Law Keeps Particle Clustering Within 10%, but Not the Gravitating Matter Spectrum at 512³

Working note

The authors — 2026-10-10, version 2.2 (v1.0 stated the hypothesis; v2.0 added the simulations; v2.1 corrected the wording of the control and the inputs after an internal audit, numbers unchanged; v2.2 corrects the growth claim: the numbers were measured on the particle field, and on the gravitating field the rule fails at 512³) — AI-assisted research programme; not peer reviewed.

Abstract

We study the law $a_0 = \kappa c \sqrt{G \rho_\Lambda}$ ($\kappa = \frac{1}{2}$, fitted) with the kernel $\nu(y) = 1/(1 - e^{-\sqrt{y}})$ and a cold fluid of fixed cosmic amount. In particle-mesh simulations, letting the law act wherever matter has collapsed makes small-scale structure too clumpy: the excess in the particle power spectrum at $k \leq 1 \text{ h Mpc}^{-1}$ is +15.3%/+19.3% on the two footings at 256³ and +17.4% at 512³, against an allowed 10%. We test one rule with no tuned radius. The law’s extra (phantom) mass is the cold fluid itself, so a halo can build the phantom only until its own cold fluid is used up; for an undepleted halo this happens at $r_{\text{edge}} = r_M / \ln[1/(1 - f_b)] \approx 5.85 r_M$. The cold fluid used is drawn from the same halo’s turnaround sphere, so mass is conserved halo by halo. With this rule the excess in the particle (baryon-tracing) power spectrum is 2.7% at 256³ and 3.3% at 512³, with σ_8 within 0.6% of a control run without the law (the cold fluid evolved as ordinary cold matter, i.e. Λ CDM-equivalent growth). It holds in three random realisations, on both footings, with a_0 tracking dark energy, and when the two settings inherited from earlier work are varied, on the particle density; on the gravitating density it holds at 256³ only. In the gravitating matter density, which includes the cold fluid the rule settles into each halo and is what lensing measures, the excess is 6.3–8.2% at 256³ but 16–19% at 512³ ($\sigma_8 + 3\text{--}5\%$), above the 10% limit, in every 512³ run. Turning the mass conservation off restores the excess (15.4% in the particle spectrum). Earlier rules that set the edge by hand, or from measured baryon censuses, sit at the 10% limit at 512³. The rule adds no tuned radius or width; its framework inputs are κ and the cosmic baryon fraction, and it inherits standard ingredients (a Λ CDM initial power spectrum, a spherical-collapse turnaround density, a halo finder and the pass thresholds). The cold fluid is handled as bookkeeping, not as simulated particles, and κ , the cold fluid’s amount and ρ_Λ remain undetermined.

1 Setting

The acceleration scale is $a_0 = \kappa c \sqrt{G \rho_\Lambda}$ with $\kappa = \frac{1}{2}$ fitted. We report two footings, 9.3603×10^{-11} and $1.1312 \times 10^{-10} \text{ m s}^{-2}$, separately. The phantom density is $\rho_{\text{ph}} = \nabla \cdot [(\nu - 1)g_b]/4\pi G$. A cold fluid with the cosmic amount $\Omega_c/\Omega_b = 5.364$ is assumed; no particle species is specified and the amount is an input. In the working model the phantom is not extra mass from nowhere: it is cold fluid that has settled into the profile the law requires.

2 The problem

Particle-mesh runs: box $200 h^{-1} \text{ Mpc}$, initial redshift 49, 150 steps, the same initial conditions as a control run without the law, in which the cold fluid is evolved as ordinary cold matter (so the control has Λ CDM-equivalent growth). Pass means $|\sigma_8 - 1| \leq 5\%$ and $\max |P - 1| \leq 10\%$ for $k \leq 1 \text{ h Mpc}^{-1}$, both relative to the control.

rule	σ_8 ratio	$\max P - 1 $
phantom added as extra gravity (256 ³)	1.2049 / 1.2573	
reservoir rule with gas filter (256 ³)	1.0193 / 1.0243	0.153 / 0.193
the same at 512 ³ (canonical)	1.0226	0.174

The excess grows with resolution. In the 256³ snapshots 0.805 of the switched-on mass in filament-like cells (about 81%) lies inside resolved halos’ turnaround spheres, so the excess is a halo-outskirt effect. Galaxy lensing (KiDS-1000 isolated lenses, with a free two-halo amplitude) only requires the law out to $x = 0.48$ (canonical) and 0.53 (alternative) of the turnaround radius, and in our snapshots 0.83 of the excess source lies beyond $x = 0.4$.

3 The rule

For a point mass with this kernel the enclosed phantom is $M_{\text{ph}}(< r) = M_b/(e^s - 1)$ with $s = r_M/r$ and $r_M = \sqrt{GM_b/a_0}$. If the phantom is the halo’s own cold fluid, it can grow only until it equals that cold mass. For a halo that has kept its cosmic share of baryons this gives

$$r_{\text{edge}} = \frac{r_M}{\ln[1/(1 - f_b)]} = 5.85 r_M,$$

and the general form with retained fraction f_{ret} is $r_M/\ln(1 + f_{\text{ret}} f_b/(1 - f_b))$ (equal to $5.850 r_M$ for $f_{\text{ret}} = 1$). Version 1.0 used measured retention ($f_{\text{ret}} = 0.07\text{--}0.10$), which places the edge at 0.24–0.54 of the turnaround radius for KiDS lenses and 0.24–0.33 for the simulated groups and clusters. The simulation does not remove baryons, so the self-consistent choice there is $f_{\text{ret}} = 1$.

In the simulation, inside each halo’s edge the excess of the phantom over the local cold density is added as a source. The same amount is then removed from the cold fluid inside that halo’s turnaround sphere (in proportion to where the cold fluid is), so the added source sums to zero on every catchment. Nothing in this rule is tuned: there is no confinement radius and no catchment width. Two settings are inherited from earlier lanes: the width of the collapse switch and a gas-phase filter on the baryon source. We vary both below.

4 Results

run	σ_8 ratio	$\max P - 1 $
256 ³ , canonical / alternative	1.0033 / 1.0031	0.027 / 0.029
256 ³ , two more realisations (canonical)	1.0024 / 1.0026	0.022 / 0.017
256 ³ , the same two (alternative)	1.0023 / 1.0030	0.025 / 0.023
256 ³ , a_0 tracking dark energy	1.0035 / 1.0032	0.029 / 0.030
256 ³ , switch width halved / doubled	1.0033 / 1.0033	0.027 / 0.027
256 ³ , other gas filters	1.0027 / 1.0038	0.026 / 0.031
512³ , canonical	1.0045	0.033
512 ³ , alternative	1.0054	0.040
512 ³ , a_0 tracking dark energy	1.0045	0.033
control: mass conservation switched off (256 ³)	1.0326	0.154
edge set by hand at $0.4 r_{\text{ta}}$ (512 ³)	1.0046 / 1.0059	0.080 / 0.0996
edge from census retention (512 ³)	1.0072 / 1.0073	0.092 / 0.1009

On the particle density every run of the rule passes, 15 of 15 (on the gravitating density 12 of 15 pass; see the correction below). The excess changes from 2.7% to 3.3% between 256³ and 512³, a much smaller rise than the earlier rules showed; with two resolutions and one 512³ realisation, convergence is not established. No catchment is ever asked for more cold fluid than it holds. The largest share of a catchment’s cold fluid used is 30% at 256³ and 57% at 512³; that share rises with resolution and is the quantity to watch at higher resolution. The control shows that the mass conservation, not the edge alone, removes the excess from the particle spectrum.

Correction (2026-10-10). The spectra in the table above, and in all earlier versions of this note, were measured on the particle density. The rule moves cold fluid within each catchment, and that moved mass gravitates, so the matter power spectrum is that of the particles plus the settled source. We re-measured every run of the rule on that density from the same saved $z = 0$ states, using each run’s own force calculation (lane CFG555; the particle numbers are reproduced to better than 10^{-7}). The second 512³ realisation, still running in v2.1, is included. Same control, same cuts.

run	particle: σ_8 / $\max P - 1 $	gravitating: σ_8 / $\max P - 1 $
256 ³ , canonical	1.0033 / 0.027	1.0158 / 0.075
256 ³ , alternative	1.0031 / 0.029	1.0156 / 0.077
256 ³ , realisation 2 (canonical)	1.0024 / 0.022	1.0141 / 0.063
256 ³ , realisation 3 (canonical)	1.0026 / 0.017	1.0175 / 0.075
256 ³ , realisation 2 (alternative)	1.0023 / 0.025	1.0149 / 0.068
256 ³ , realisation 3 (alternative)	1.0030 / 0.023	1.0193 / 0.082
256 ³ , a_0 tracking dark energy (canonical)	1.0035 / 0.029	1.0157 / 0.074
256 ³ , a_0 tracking dark energy (alternative)	1.0032 / 0.030	1.0154 / 0.075
256 ³ , switch width halved	1.0033 / 0.027	1.0158 / 0.075
256 ³ , switch width doubled	1.0033 / 0.027	1.0159 / 0.075
256 ³ , other gas filter (1)	1.0027 / 0.026	1.0153 / 0.072
256 ³ , other gas filter (2)	1.0038 / 0.031	1.0160 / 0.076
512³, canonical	1.0045 / 0.033	1.0405 / 0.163
512 ³ , alternative	1.0054 / 0.040	1.0459 / 0.193
512 ³ , a_0 tracking dark energy	1.0045 / 0.033	1.0429 / 0.167
512 ³ , realisation 2 (canonical)	1.0037 / 0.040	1.0333 / 0.184
control: mass conservation switched off (256 ³)	1.0326 / 0.154	1.1730 / 0.505

On the gravitating density all twelve 256³ runs pass ($\max |P - 1| = 0.063$ – 0.082 , σ_8 ratio 1.014–1.019). All four 512³ runs fail the spectrum cut (0.163–0.193 against 0.10), although σ_8 stays inside its 5% cut. At 512³ the gravitating spectrum is also 13–20% below the control at $k = 1 \, h \, \text{Mpc}^{-1}$, so the miss is a reshaping on both sides of the cut, not a single bin. The excess comes mainly from the settled cold fluid being positively correlated with the particles on catchment scales, and it grows with resolution together with the catchment draw noted above. The rows of the problem table and the two comparison rows (edge set by hand, edge from census retention), and the +15.3%/+19.3%/+17.4% numbers in the abstract, are also particle-field numbers; they were not re-measured.

5 What this does and does not show

Within this framework, the rule keeps the particle (baryon-tracing) spectrum within 10% of ΛCDM -equivalent growth at 256³ and 512³, with no radius or width chosen to fit. It does not keep the gravitating matter spectrum within 10% at 512³, and the gap grows with resolution. Structure growth as seen by lensing (weak lensing, cosmic shear S_8 , cluster counts) is therefore not fixed by this rule; galaxy clustering, which traces the baryons, corresponds to the particle field up to bias. The rule’s framework inputs are κ (fitted to rotation curves) and f_b ; it also inherits a ΛCDM initial spectrum, a spherical-collapse turnaround density, a halo finder and the pass thresholds.

It does not show the following:

- The cold fluid is bookkeeping in these runs. Its settling into the phantom profile is imposed, not derived from an action.
- Two random realisations were run at 512³; both fail on the gravitating density. The mesh does not resolve galaxy scales, and the share of cold fluid used (57% at 512³) could reach 100% at higher resolution.
- The pass criteria compare with a control that has ΛCDM -equivalent growth, not with survey data. A pass means the law does not spoil standard growth by more than 10%.
- κ , the cold fluid’s amount and ρ_Λ are not derived, and no dark-matter particle is identified.

6 Reproducibility

Each test’s pass rule was committed before the run, and each test includes a deliberately broken control. The lanes are CFG414, CFG416 and CFG422–CFG427, CFG425, CFG439 and CFG460, and the gravitating-field re-measurement is CFG555, in the repository’s `campaign_fresh_gravity` directory; the engine is `CFG424_turnaround_catchment/cfg424_pm.py` (mode RC = 0).